Supervised Baseline Evaluation on Spectrogram Images for BCI Competition IV-2a

Task 2 Report

Course Code : CSE 475
Course Title : Machine Learning
Section : 02
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Submitted By

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1. Task Objective

Task 2 assesses the supervised baseline models using the spectrogram image dataset generated in the Kaggle notebooks. The analysis for each of the train-test split is presented separately in this report for ease of comparability. Metrics that have been outputted in the notebook are the only ones provided for each of the available splits and include Accuracy, Macro Precision, Macro Recall, Macro F1 score, ROC AUC, training computation time, testing computation time, and GFLOPs per inference (with a fixed input size of 224 x 224).

2. Experimental Setting

The experiment employs 2,583 spectrogram images derived from BCI Competition IV Dataset 2a. The target classes consist of Feet, Left Hand, Right Hand, and Tongue. There are five baseline models that are tested in a supervised fashion and pre-trained on ImageNet: EfficientNet-B0, MobileNetV2, ResNet50, ViT-Tiny, and ConvNeXt-Tiny. In the revised report, all split results that have been uploaded are included: 10:90, 20:80, 30:70, 40:60, 50:50, 60:40, 70:30, 80:20, and 90:10.

Item	Value
Dataset	BCI Competition IV Dataset 2a
Total images reported by code	2,583
Classes	Feet, Left Hand, Right Hand, Tongue
Supervised input size	224 x 224
Models	EfficientNet-B0, MobileNetV2, ResNet50, ViT-Tiny, ConvNeXt-Tiny
Available splits	10:90, 20:80, 30:70, 40:60, 50:50, 60:40, 70:30, 80:20, 90:10

Table 1. Experimental setting used in the supervised baseline report.

3. Results for the 10:90 Split

For the 10:90 configuration, the dataset was divided into 232 training images, 26 validation images, and 2325 test images in the dataset. The table below presents the supervised Task 2 metrics that can be found in the notebook output.

Model	Accuracy	Precision	Recall	F1	ROC-AUC	Train(s)	Test(s)	GFLOPs
ViT-Tiny	0.2624	0.2668	0.2618	0.2217	0.5023	66.3	4.4930	2.1493
EfficientNet-B0	0.2495	0.2381	0.2494	0.1899	0.4982	69.2	4.6120	0.8277
MobileNetV2	0.2452	0.2407	0.2458	0.2023	0.4986	66.5	4.3900	0.6524
ConvNeXt-Tiny	0.2370	0.1185	0.2376	0.1580	0.4874	147.9	12.7450	8.9096
ResNet50	0.2366	0.2446	0.2367	0.2211	0.4833	104.2	7.9970	8.2634

Table 2. Supervised Task 2 results for the 10:90 split.

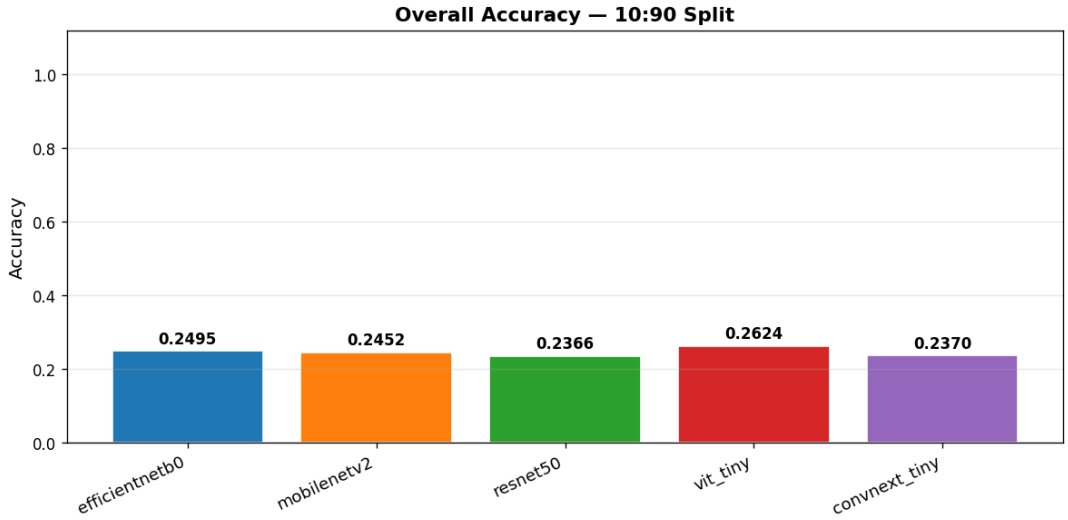

Figure 1. Overall accuracy comparison for the 10:90 split.

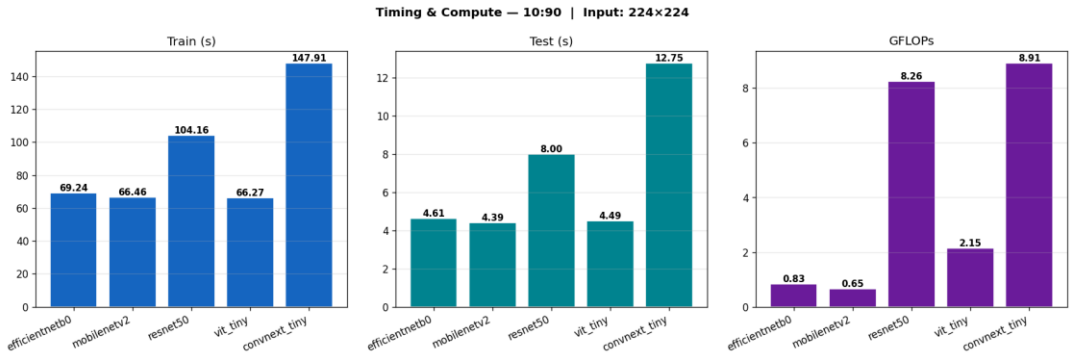

Figure 2. Training time, testing time, and GFLOPs comparison for the 10:90 split.

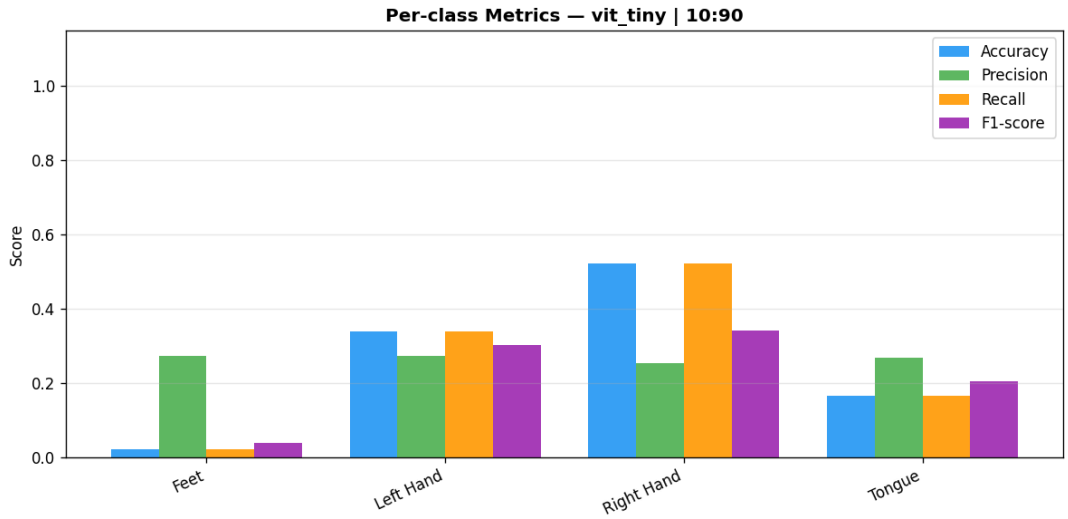

Figure 3. Per-class accuracy, precision, recall, and F1-score for the best-performing model in the 10:90 split (ViT-Tiny).

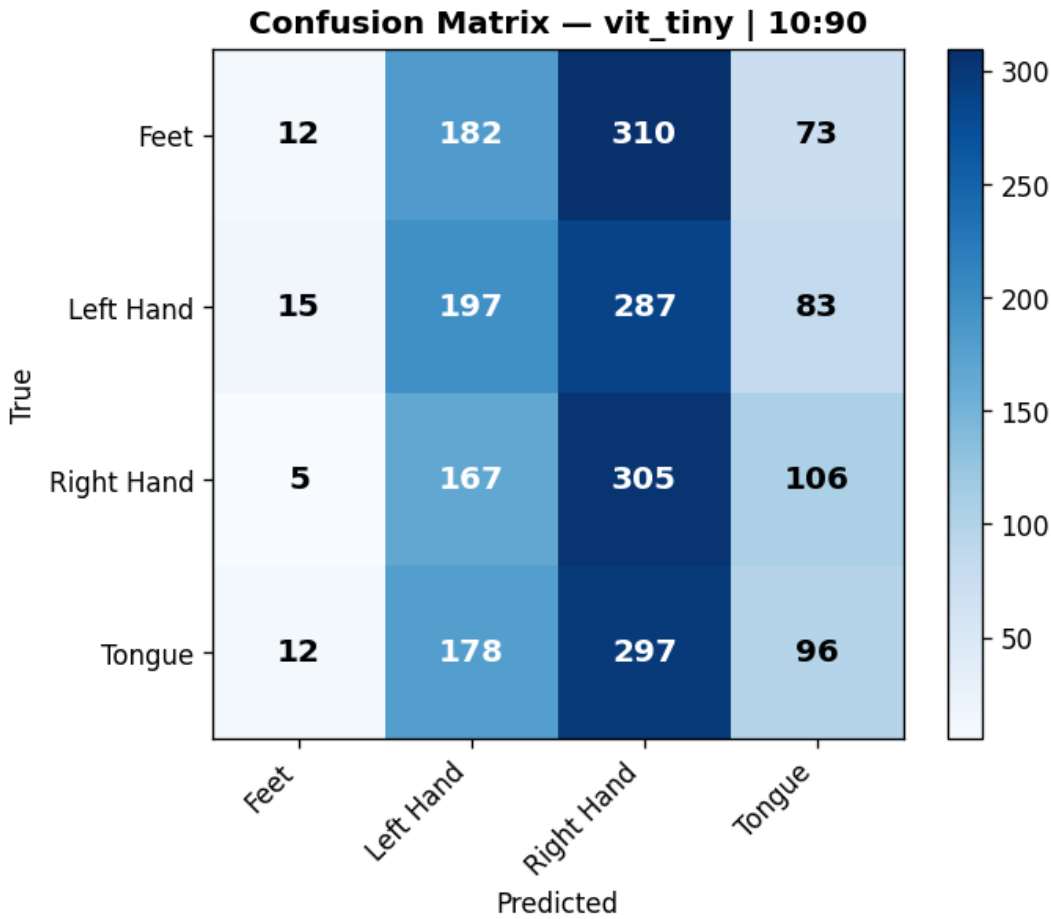

Figure 4. Confusion matrix for the best-performing model in the 10:90 split (ViT-Tiny).

The best-performing supervised model for the 10:90 split is ViT-Tiny, which has an accuracy of 0.2624, macro F1 score of 0.2217, and ROC-AUC of 0.5023. The best model for this split is ViT-Tiny. The confusion matrix demonstrates that the model overpredicts the Right Hand class. Some of the mistakes made by the model include Feet -> Right Hand, Tongue -> Right Hand, and Left Hand -> Right Hand.

4. Results for the 20:80 Split

For the 20:80 configuration, the dataset contained 464 training samples, 52 validation samples, and 2067 test samples. Below is the table that contains the results obtained from the supervised Task 2 only using the notebook output.

Model	Accuracy	Precision	Recall	F1	ROC-AUC	Train(s)	Test(s)	GFLOPs
ResNet50	0.2675	0.2662	0.2670	0.2433	0.5169	177.5	7.0920	8.2634
EfficientNet-B0	0.2564	0.2571	0.2568	0.2355	0.5034	108.9	4.2190	0.8277
ConvNeXt-Tiny	0.2511	0.0628	0.2500	0.1003	0.4956	264.8	11.6330	8.9096
ViT-Tiny	0.2501	0.1868	0.2499	0.2112	0.4988	99.4	3.8840	2.1493
MobileNetV2	0.2467	0.2766	0.2479	0.1634	0.4936	98.2	3.8530	0.6524

Table 3. Supervised Task 2 results for the 20:80 split.

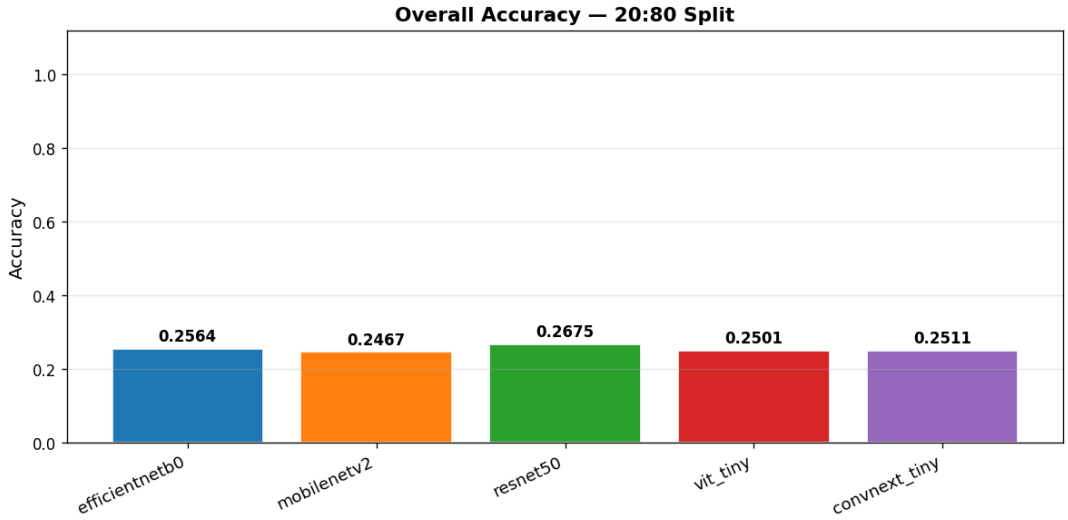

Figure 5. Overall accuracy comparison for the 20:80 split.

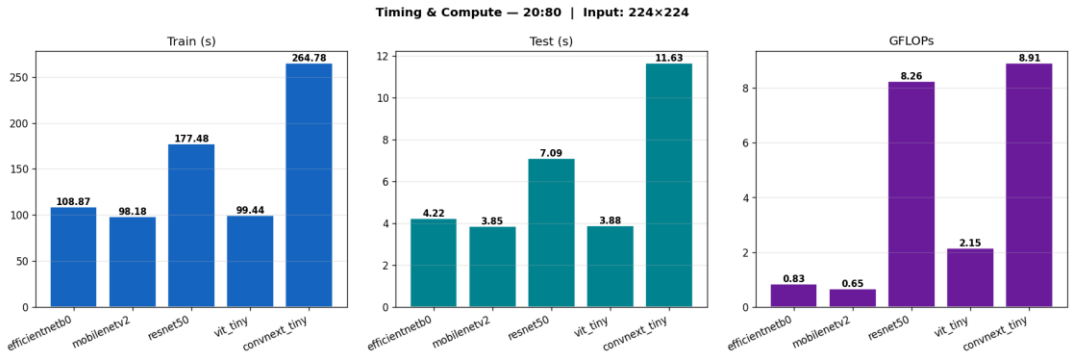

Figure 6. Training time, testing time, and GFLOPs comparison for the 20:80 split.

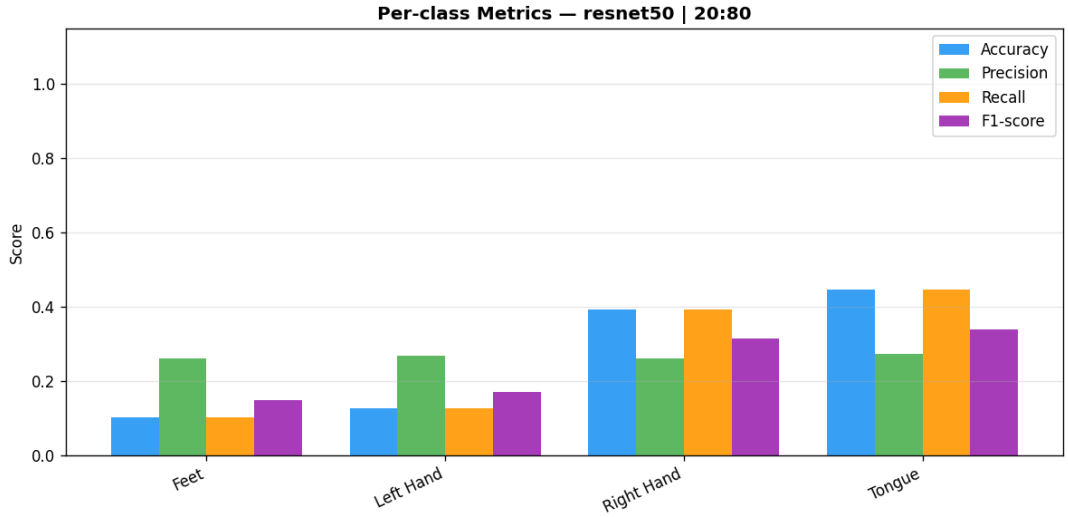

Figure 7. Per-class accuracy, precision, recall, and F1-score for the best-performing model in the 20:80 split (ResNet50).

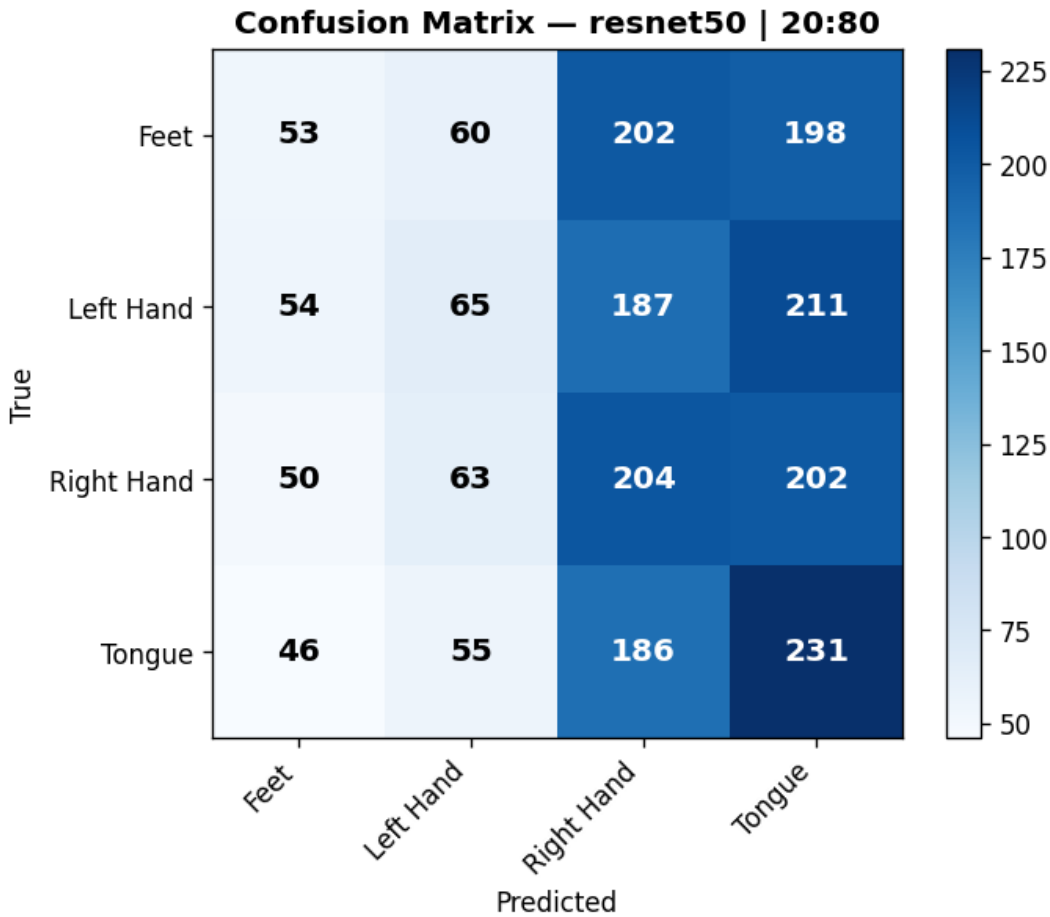

Figure 8. Confusion matrix for the best-performing model in the 20:80 split (ResNet50).

The best-performing supervised model for the 20:80 split is ResNet50, with accuracy = 0.2675, macro F1-score = 0.2433, and ROC-AUC = 0.5169. The best model for this split is ResNet50. The confusion matrix shows substantial confusion toward the Tongue and Right Hand classes. The largest errors include Left Hand -> Tongue, Right Hand -> Tongue, Feet -> Right Hand, and Feet -> Tongue.

5. Results for the 30:70 Split

For the 30:70 partition, the data set consists of 696 images for training, 78 images for validation, and 1809 images for testing. Below is a table that captures only the supervised Task 2 results from the output notebook.

Model	Accuracy	Precision	Recall	F1	ROC-AUC	Train(s)	Test(s)	GFLOPs
ConvNeXt-Tiny	0.2631	0.1952	0.2623	0.2025	0.5137	568.3	10.7840	8.9096
ViT-Tiny	0.2559	0.3175	0.2556	0.2081	0.5104	143.4	3.4410	2.1493
EfficientNet-B0	0.2548	0.2346	0.2540	0.1598	0.5075	149.3	3.6250	0.8277
MobileNetV2	0.2499	0.2475	0.2498	0.2300	0.4972	188.5	3.3560	0.6524
ResNet50	0.2471	0.2468	0.2471	0.1751	0.5002	261.5	6.4440	8.2634

Table 4. Supervised Task 2 results for the 30:70 split.

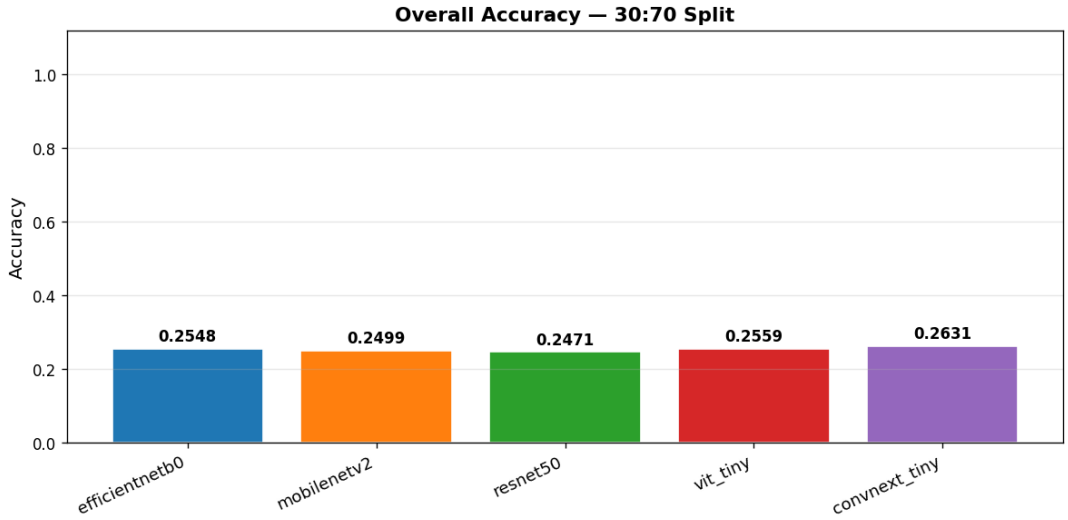

Figure 9. Overall accuracy comparison for the 30:70 split.

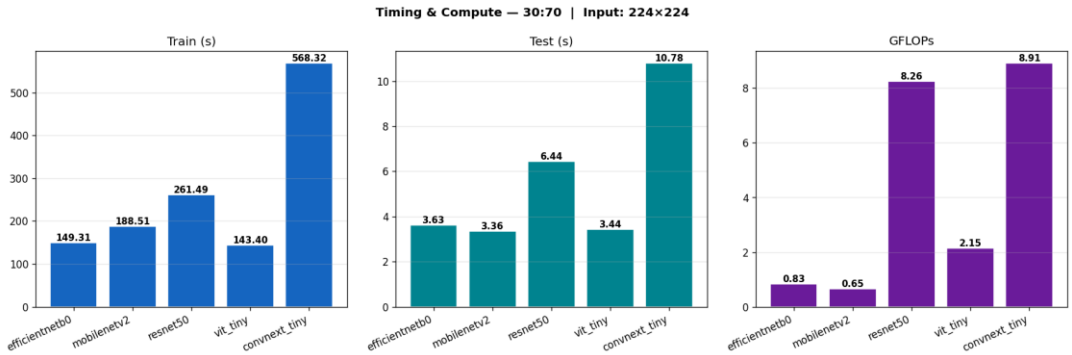

Figure 10. Training time, testing time, and GFLOPs comparison for the 30:70 split.

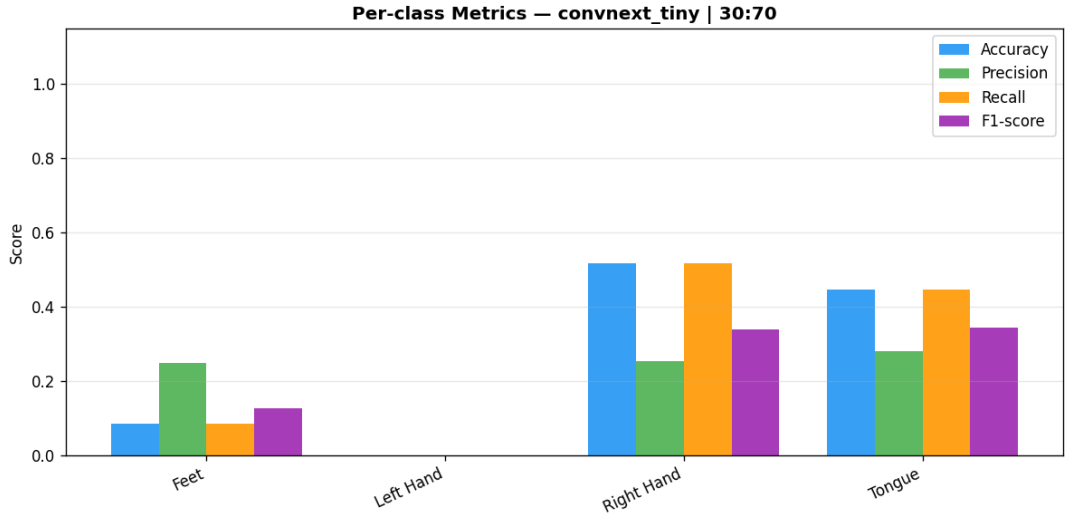

Figure 11. Per-class accuracy, precision, recall, and F1-score for the best-performing model in the 30:70 split (ConvNeXt-Tiny).

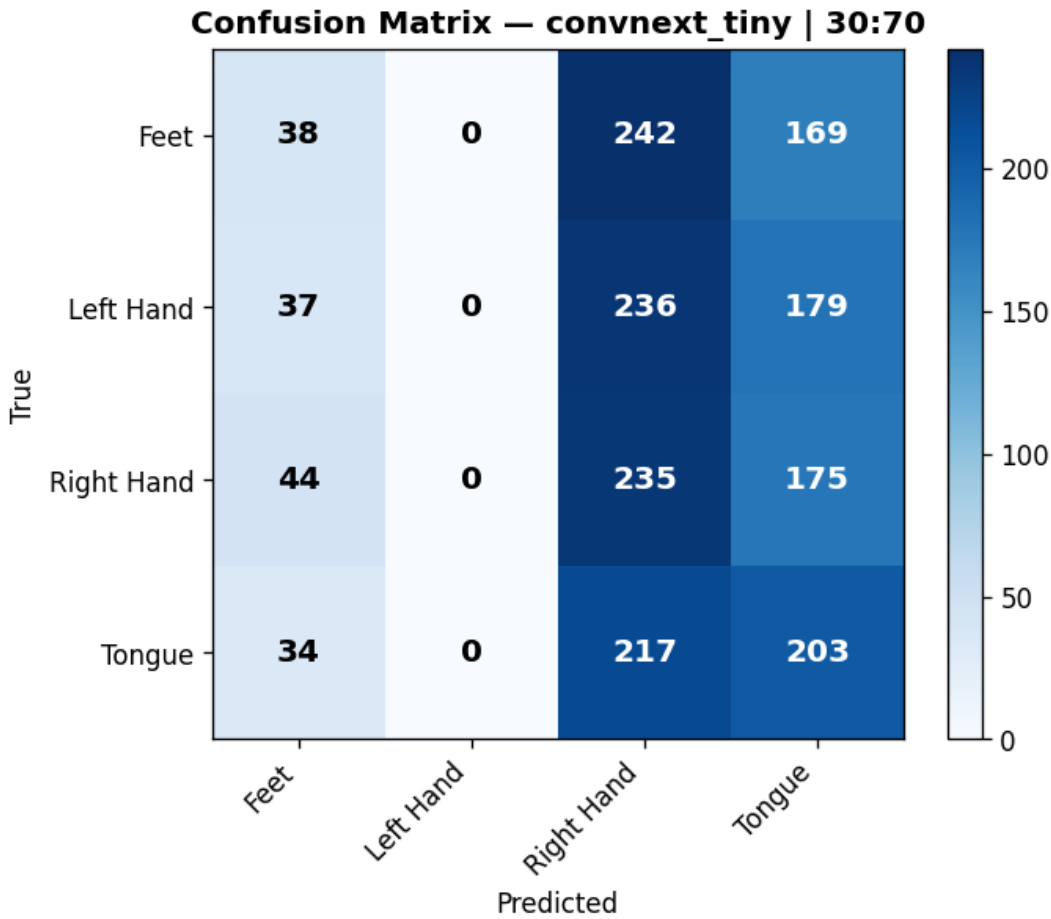

Figure 12. Confusion matrix for the best-performing model in the 30:70 split (ConvNeXt-Tiny).

The best-performing supervised model for the 30:70 split is ConvNeXt-Tiny, with accuracy = 0.2631, macro F1-score = 0.2025, and ROC-AUC = 0.5137. The best model for this split is ConvNeXt-Tiny. The confusion matrix indicates weak recognition of the Left Hand class and frequent predictions as Right Hand or Tongue. The main failure modes are Feet -> Right Hand, Left Hand -> Right Hand, Tongue -> Right Hand, and Left Hand -> Tongue.

6. Results for the 40:60 Split

For the 40:60 configuration, the dataset was divided into 929 training images, 104 validation images, and 1550 testing images. The table below reports only the supervised Task 2 metrics available in the notebook output for this split.

Model	Accuracy	Precision	Recall	F1	ROC-AUC	Train(s)	Test(s)	GFLOPs
ResNet50	0.2561	0.2821	0.2560	0.2182	0.4980	384.1	5.4090	8.2634
EfficientNet-B0	0.2548	0.2389	0.2547	0.1956	0.4970	194.5	3.2920	0.8277
ViT-Tiny	0.2510	0.1242	0.2500	0.1646	0.5037	189.8	3.0840	2.1493
ConvNeXt-Tiny	0.2419	0.1125	0.2410	0.1255	0.4886	479.7	9.3200	8.9096
MobileNetV2	0.2368	0.2390	0.2370	0.2310	0.4943	188.1	3.0310	0.6524

Table 5. Supervised Task 2 results for the 40:60 split.

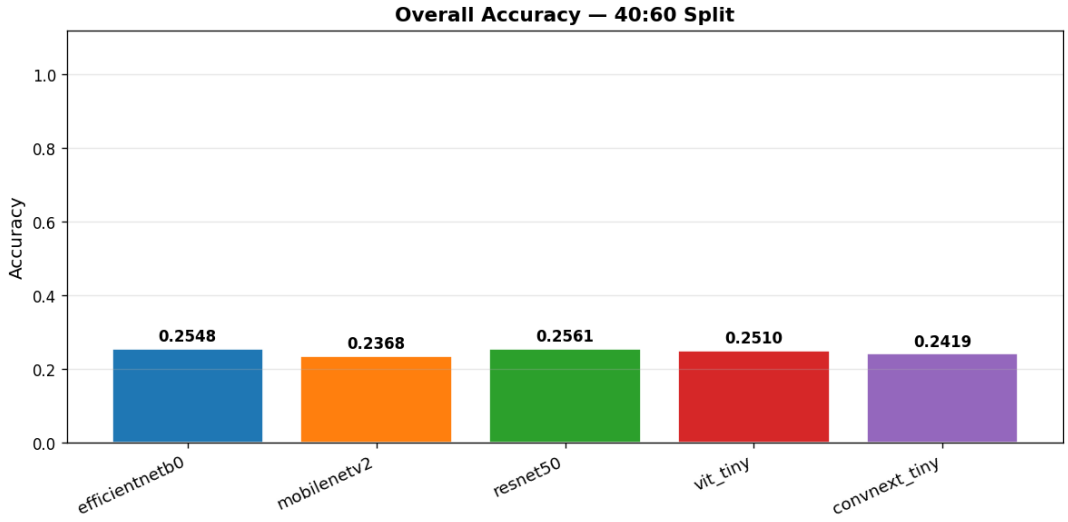

Figure 13. Overall accuracy comparison for the 40:60 split.

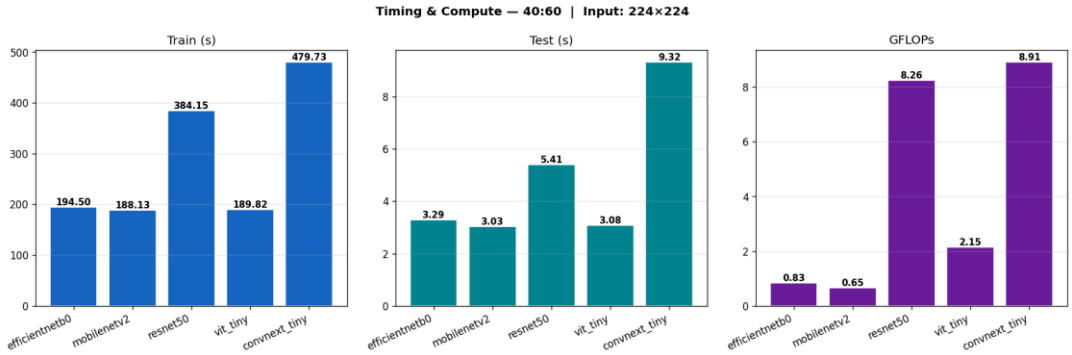

Figure 14. Training time, testing time, and GFLOPs comparison for the 40:60 split.

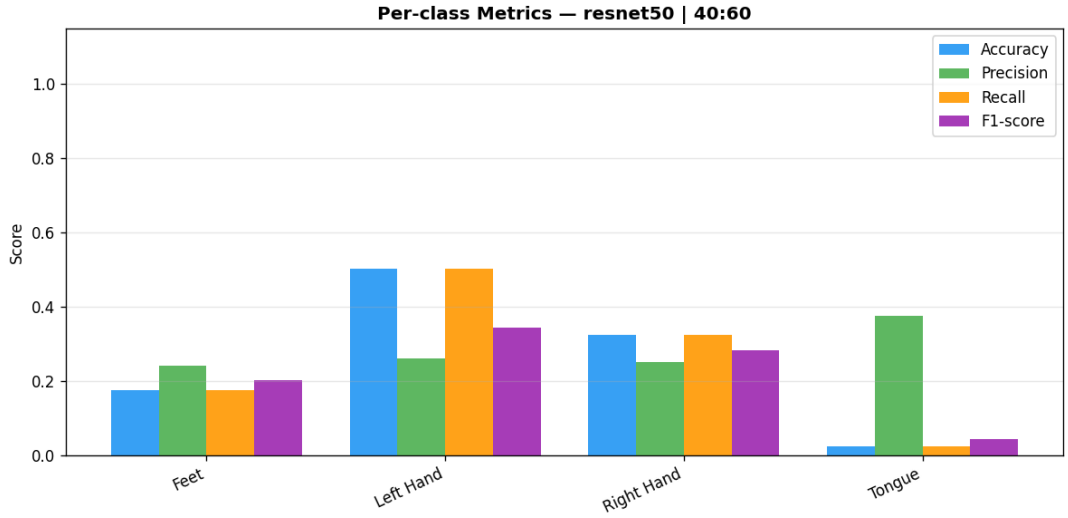

Figure 15. Per-class accuracy, precision, recall, and F1-score for the best-performing model in the 40:60 split (ResNet50).

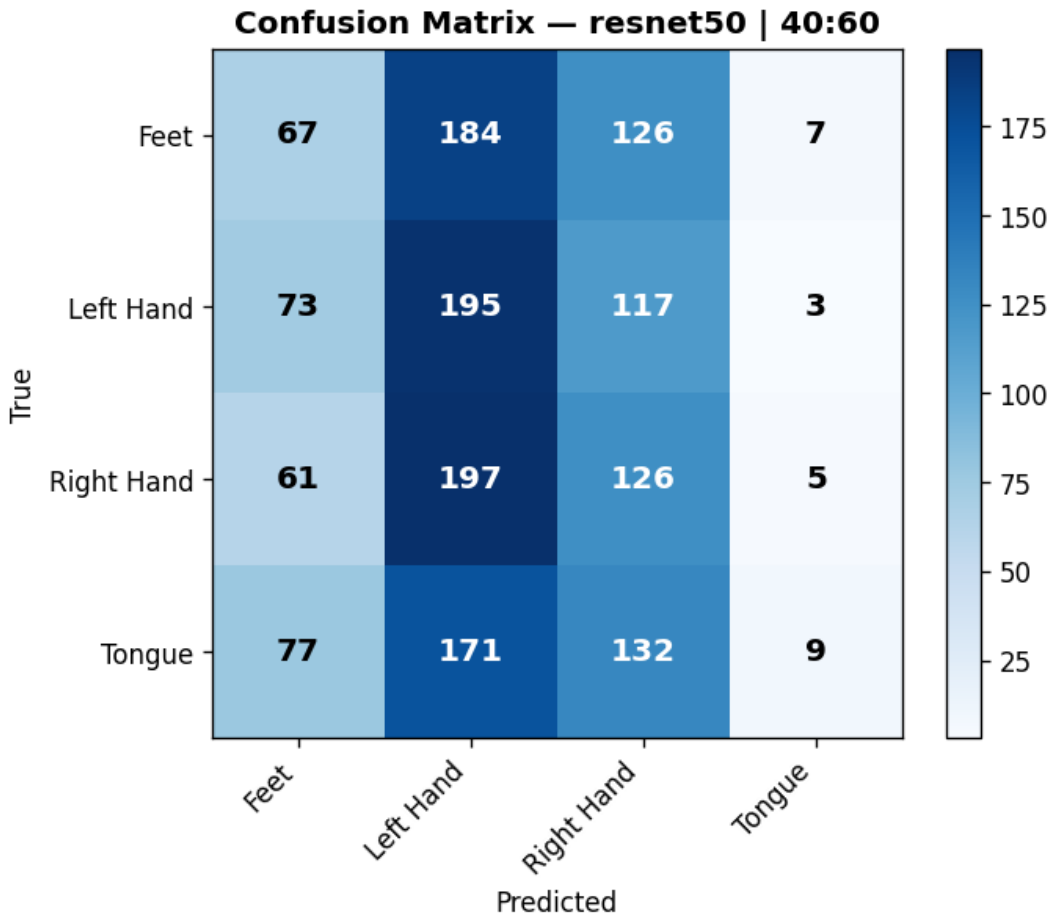

Figure 16. Confusion matrix for the best-performing model in the 40:60 split (ResNet50).

The best supervised model for the 30:70 split is ConvNeXt-Tiny, which has an accuracy of 0.2631, a macro F1 score of 0.2025, and ROC-AUC of 0.5137. The model ConvNeXt-Tiny gives the best performance on this split. The main confused pairs are Right Hand -> Left Hand, Feet -> Left Hand, and Tongue -> Left Hand, with additional confusion toward Right Hand.

7. Results for the 50:50 Split

For the 50:50 configuration, the dataset was split as follows: 1161 training, 130 validation, and 1292 testing images. The results of supervised Task 2 are presented in the table below .

Model	Accuracy	Precision	Recall	F1	ROC-AUC	Train(s)	Test(s)	GFLOPs
ResNet50	0.2771	0.2752	0.2768	0.2701	0.5273	744.2	5.1400	8.2634
ViT-Tiny	0.2755	0.1431	0.2764	0.1806	0.5193	250.4	2.6100	2.1493
EfficientNet-B0	0.2709	0.2782	0.2709	0.2649	0.5165	324.2	2.7990	0.8277
ConvNeXt-Tiny	0.2663	0.1377	0.2672	0.1697	0.5073	740.2	8.7800	8.9096
MobileNetV2	0.2523	0.2599	0.2520	0.2012	0.5064	296.0	2.6070	0.6524

Table 6. Supervised Task 2 results for the 50:50 split.

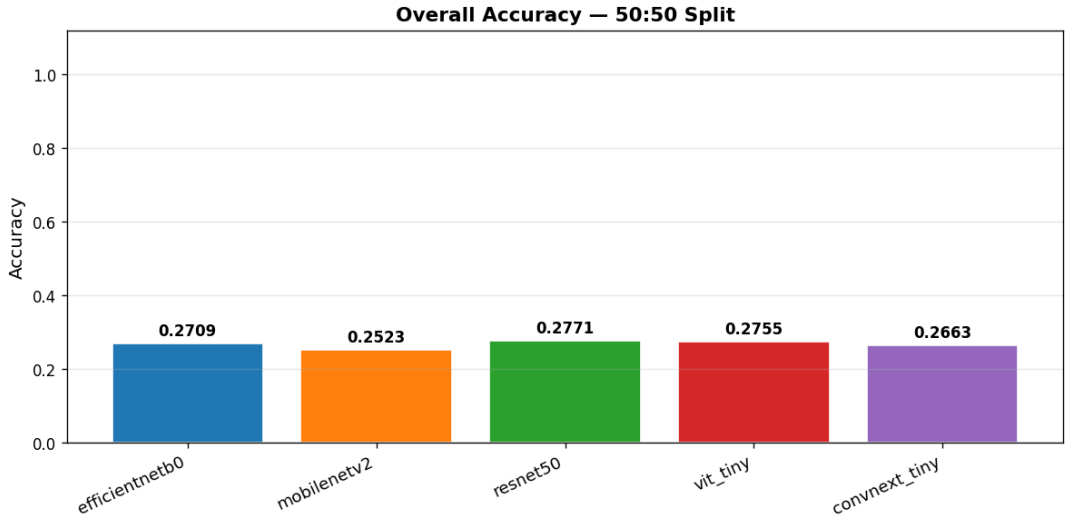

Figure 17. Overall accuracy comparison for the 50:50 split.

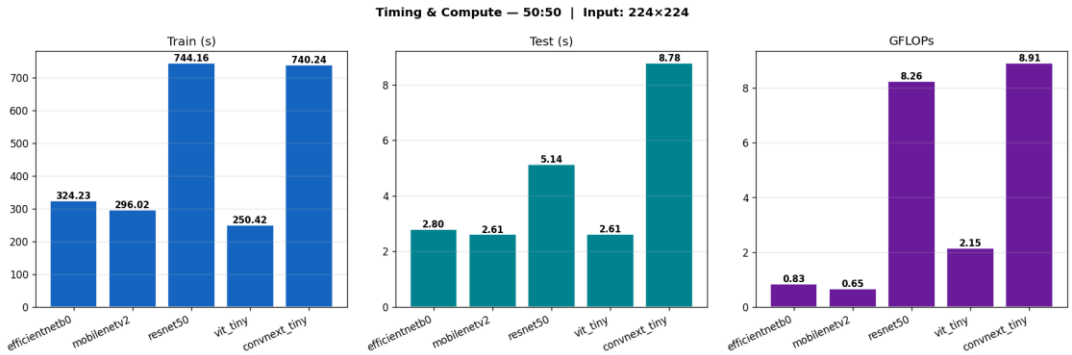

Figure 18. Training time, testing time, and GFLOPs comparison for the 50:50 split.

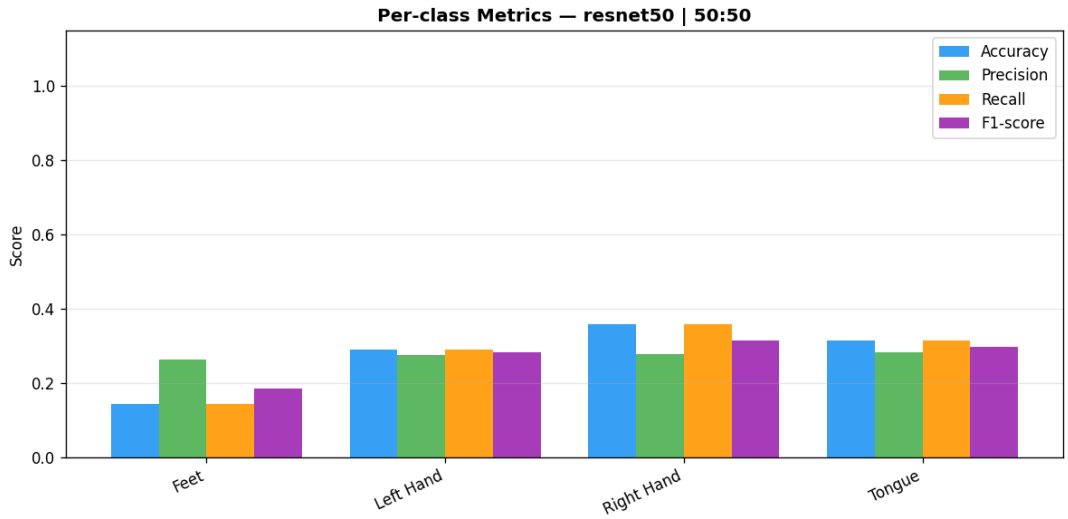

Figure 19. Per-class accuracy, precision, recall, and F1-score for the best-performing model in the 50:50 split (ResNet50).

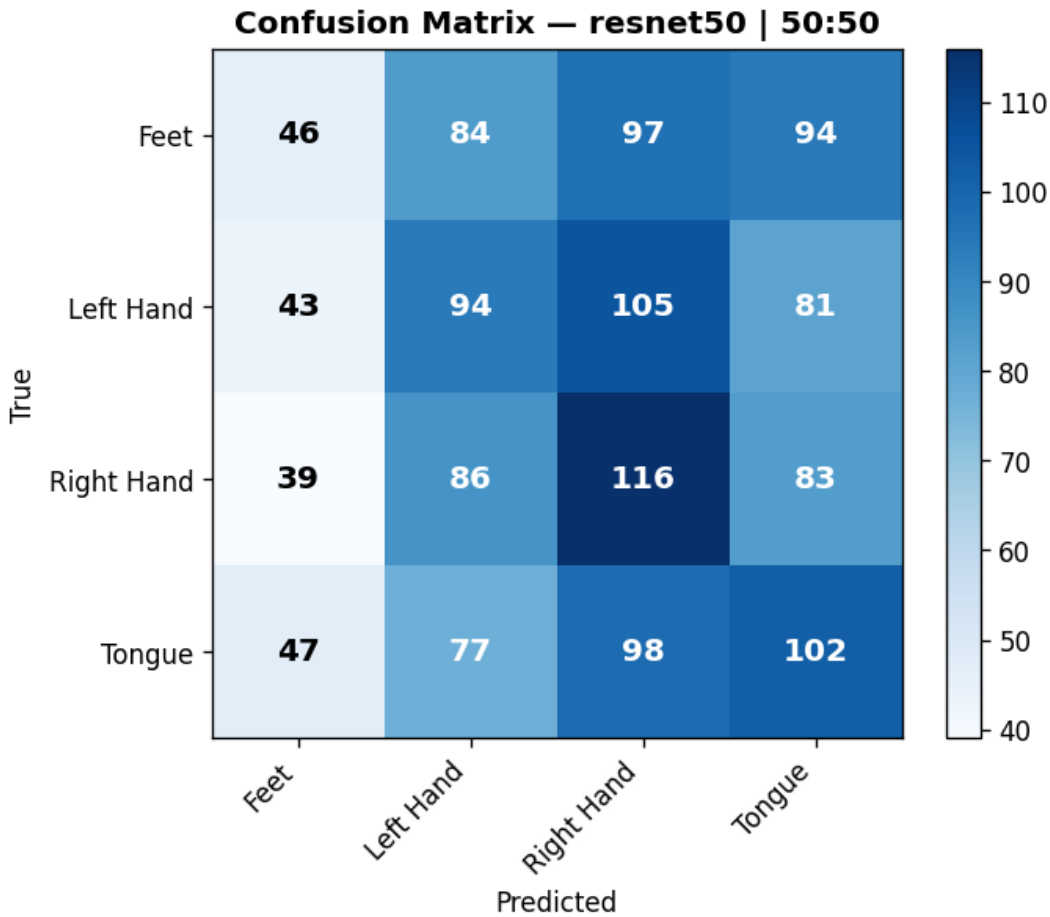

Figure 20. Confusion matrix for the best-performing model in the 50:50 split (ResNet50).

The best-performing supervised model for the 50:50 split is ResNet50, with accuracy = 0.2771, macro F1-score = 0.2701, and ROC-AUC = 0.5273. The best model for this split is ResNet50. The confusion matrix shows broad off-diagonal confusion. The most visible class confusions are Left Hand -> Right Hand and Tongue -> Right Hand, with additional mixing among Feet, Right Hand, and Tongue.

8. Results for the 60:40 Split

For the 60:40 configuration, the dataset was divided into 1394 training images, 155 validation images, and 1034 testing images. The table below reports only the supervised Task 2 metrics available in the notebook output for this split.

Model	Accuracy	Precision	Recall	F1	ROC-AUC	Train(s)	Test(s)	GFLOPs
ViT-Tiny	0.2660	0.2033	0.2654	0.1764	0.5208	264.9	2.0550	2.1493
MobileNetV2	0.2650	0.2695	0.2647	0.2237	0.5192	263.7	2.0540	0.6524
ResNet50	0.2611	0.2780	0.2621	0.1971	0.5134	767.1	4.1570	8.2634
ConvNeXt-Tiny	0.2495	0.1305	0.2490	0.1050	0.5005	784.0	7.4480	8.9096
EfficientNet-B0	0.2427	0.2466	0.2429	0.1759	0.4999	274.4	2.2220	0.8277

Table 7. Supervised Task 2 results for the 60:40 split.

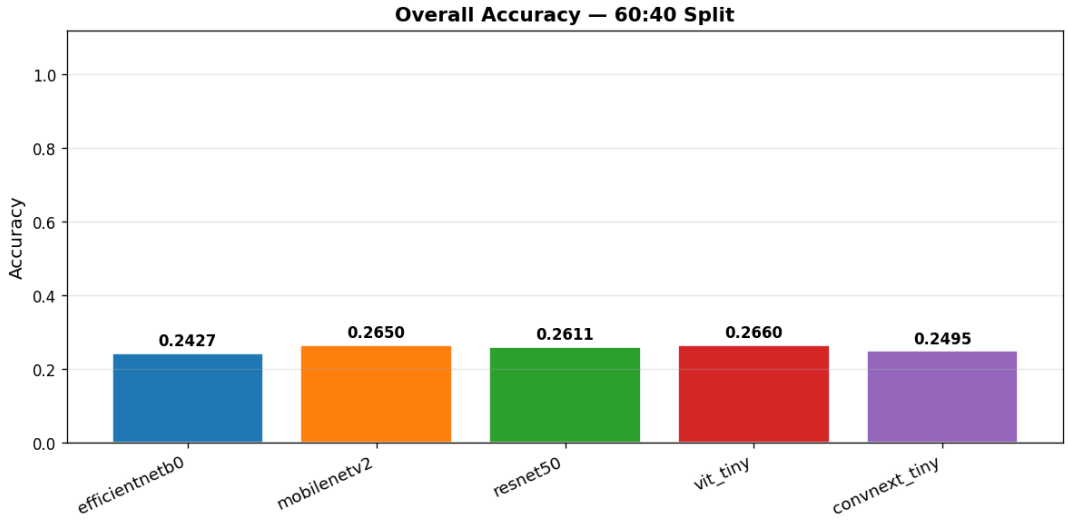

Figure 21. Overall accuracy comparison for the 60:40 split.

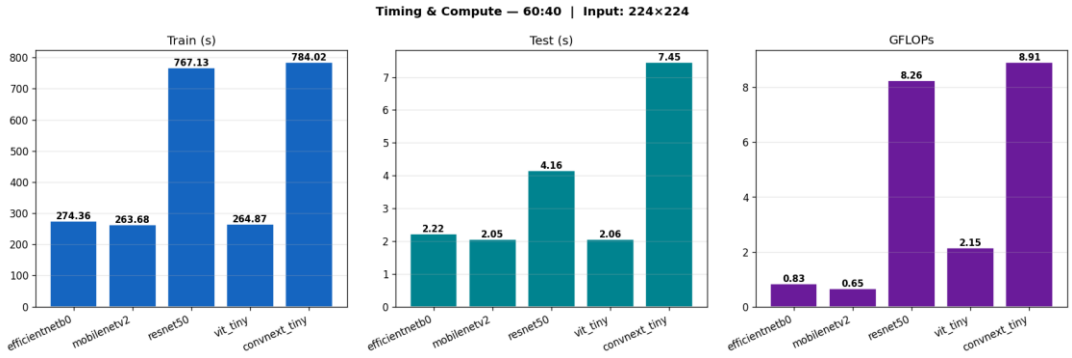

Figure 22. Training time, testing time, and GFLOPs comparison for the 60:40 split.

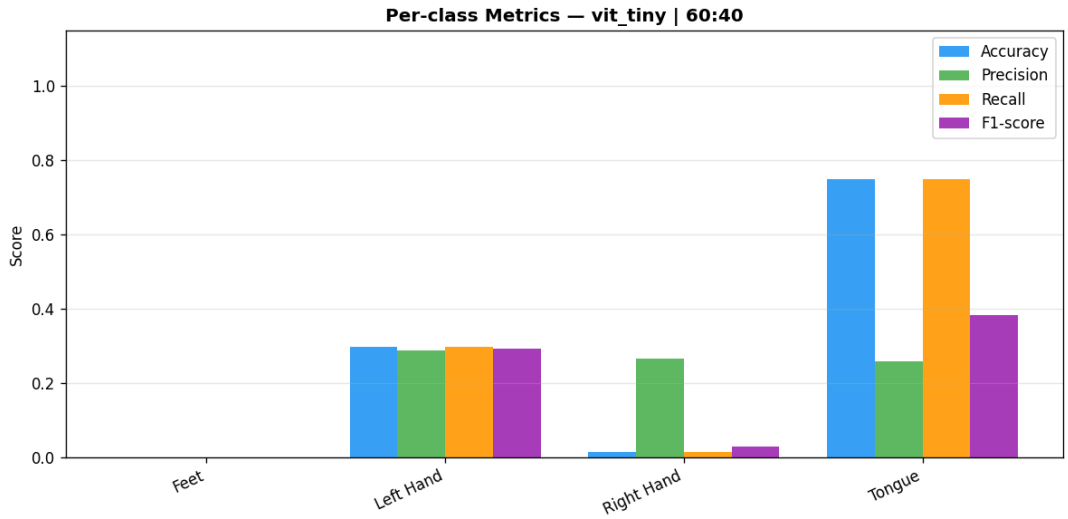

Figure 23. Per-class accuracy, precision, recall, and F1-score for the best-performing model in the 60:40 split (ViT-Tiny).

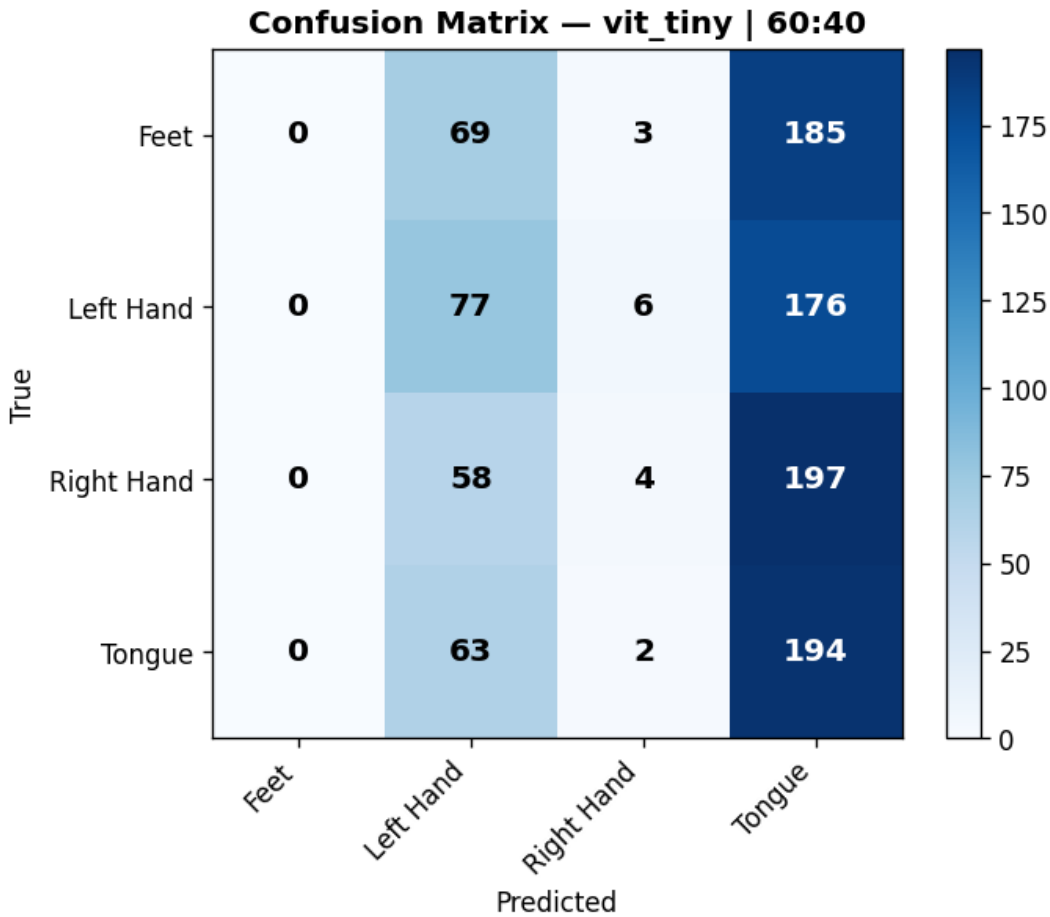

Figure 24. Confusion matrix for the best-performing model in the 60:40 split (ViT-Tiny).

The best-performing supervised model for the 60:40 split is ViT-Tiny, with accuracy = 0.2660, macro F1-score = 0.1764, and ROC-AUC = 0.5208. The best model for this split is ViT-Tiny. The confusion matrix shows a strong collapse toward the Tongue category. The largest failure modes are Right Hand -> Tongue and Feet -> Tongue, with Left Hand also frequently predicted as Tongue.

9. Results for the 70:30 Split

For the 70:30 configuration, the dataset was divided into 1627 training images, 181 validation images, and 775 testing images. The table below reports only the supervised Task 2 metrics available in the notebook output for this split.

Model	Accuracy	Precision	Recall	F1	ROC-AUC	Train(s)	Test(s)	GFLOPs
EfficientNet-B0	0.2723	0.2737	0.2722	0.2692	0.5173	361.1	1.7880	0.8277
ViT-Tiny	0.2723	0.2248	0.2712	0.1912	0.5460	304.1	1.6600	2.1493
ResNet50	0.2671	0.2831	0.2682	0.2300	0.5292	545.9	2.9250	8.2634
ConvNeXt-Tiny	0.2503	0.0626	0.2500	0.1001	0.5095	820.4	5.9240	8.9096
MobileNetV2	0.2335	0.2476	0.2335	0.2247	0.5007	300.8	1.6390	0.6524

Table 8. Supervised Task 2 results for the 70:30 split.

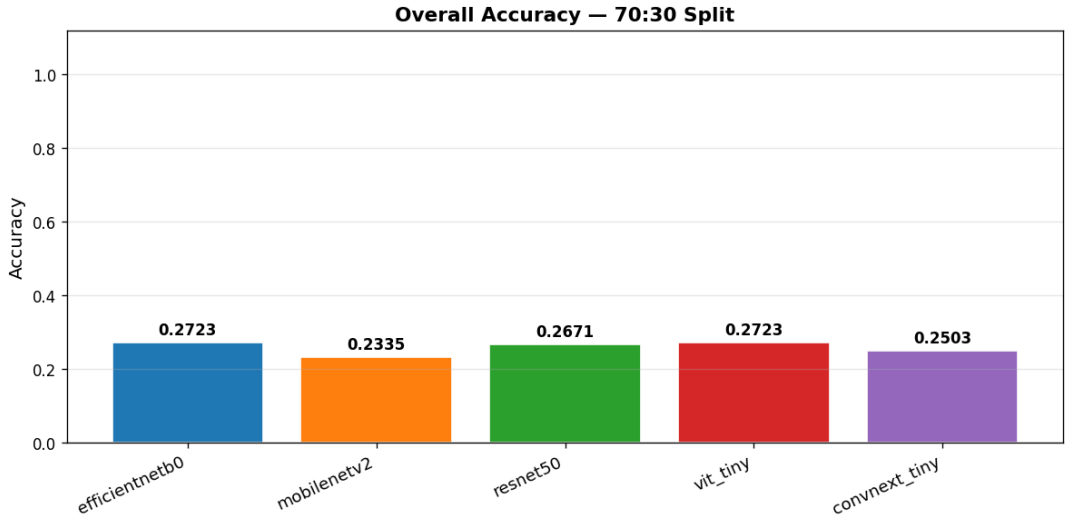

Figure 25. Overall accuracy comparison for the 70:30 split.

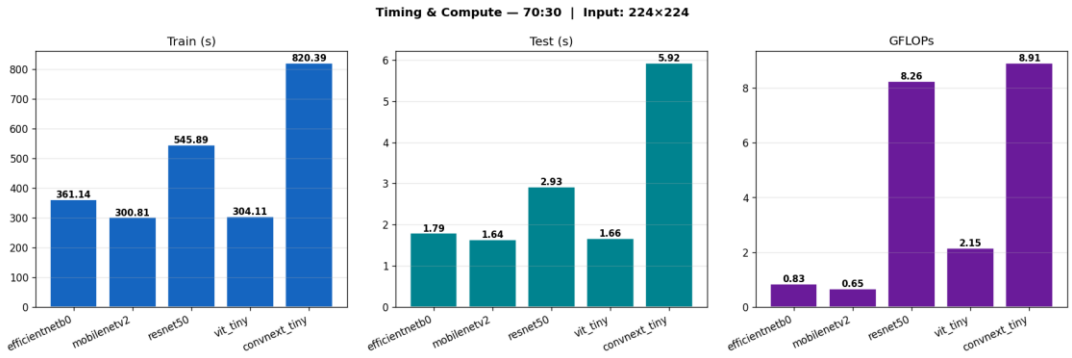

Figure 26. Training time, testing time, and GFLOPs comparison for the 70:30 split.

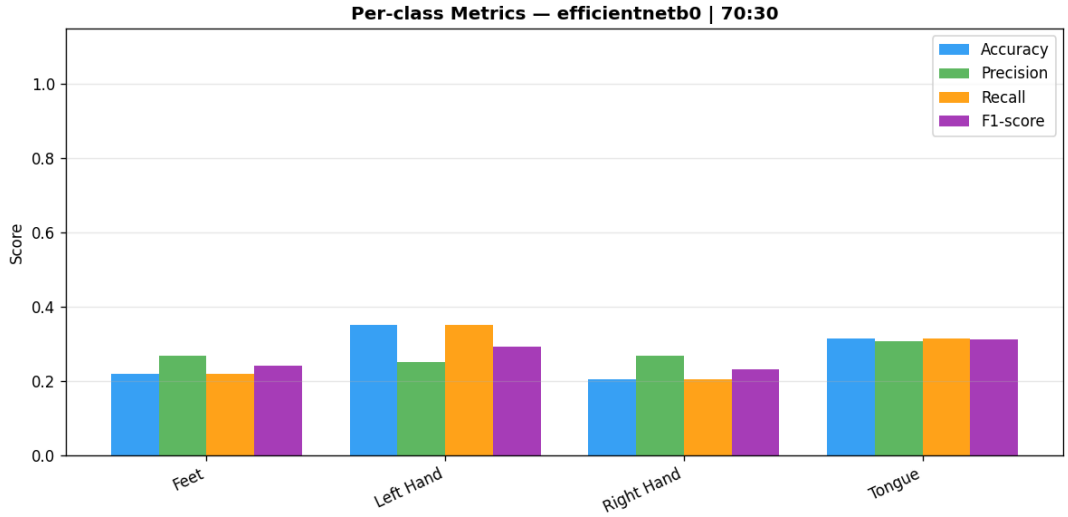

Figure 27. Per-class accuracy, precision, recall, and F1-score for the best-performing model in the 70:30 split (EfficientNet-B0).

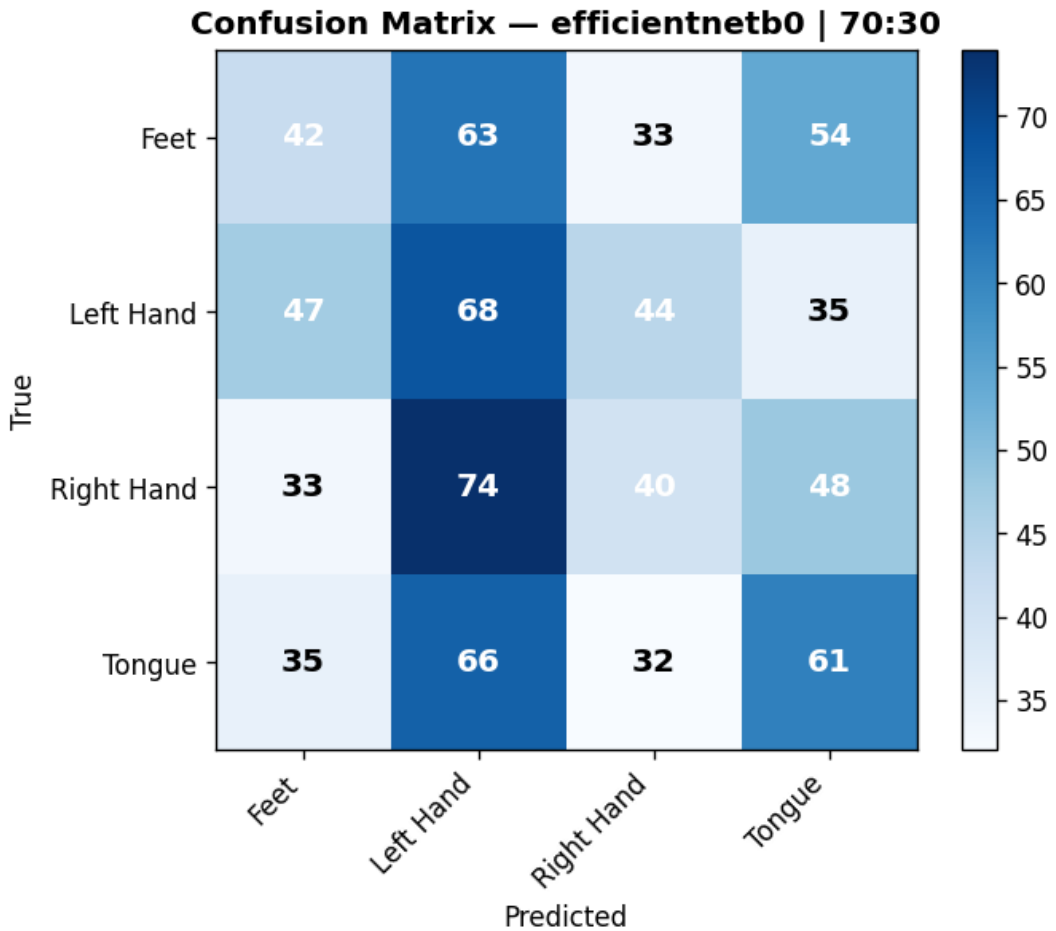

Figure 28. Confusion matrix for the best-performing model in the 70:30 split (EfficientNet-B0).

The best-performing supervised model for the 70:30 split is EfficientNet-B0, with accuracy = 0.2723, macro F1-score = 0.2692, and ROC-AUC = 0.5173. The best model for this split is EfficientNet-B0. The confusion matrix shows that the model often over-predicts Left Hand. The dominant errors are Right Hand -> Left Hand, Tongue -> Left Hand, and Feet -> Left Hand, with some Feet -> Tongue confusion.

10. Results for the 80:20 Split

For the 80:20 configuration, the dataset was divided into 1859 training images, 207 validation images, and 517 testing images. The table below reports only the supervised Task 2 metrics available in the notebook output for this split.

Model	Accuracy	Precision	Recall	F1	ROC-AUC	Train(s)	Test(s)	GFLOPs
ResNet50	0.2979	0.3068	0.2971	0.2784	0.5418	760.8	2.5540	8.2634
MobileNetV2	0.2747	0.2986	0.2743	0.2254	0.5145	368.6	1.2530	0.6524
EfficientNet-B0	0.2573	0.2481	0.2567	0.2315	0.5037	384.0	1.4540	0.8277
ViT-Tiny	0.2573	0.2376	0.2558	0.1819	0.5257	372.6	1.2360	2.1493
ConvNeXt-Tiny	0.2573	0.1344	0.2593	0.1441	0.5101	1173.1	5.4400	8.9096

Table 9. Supervised Task 2 results for the 80:20 split.

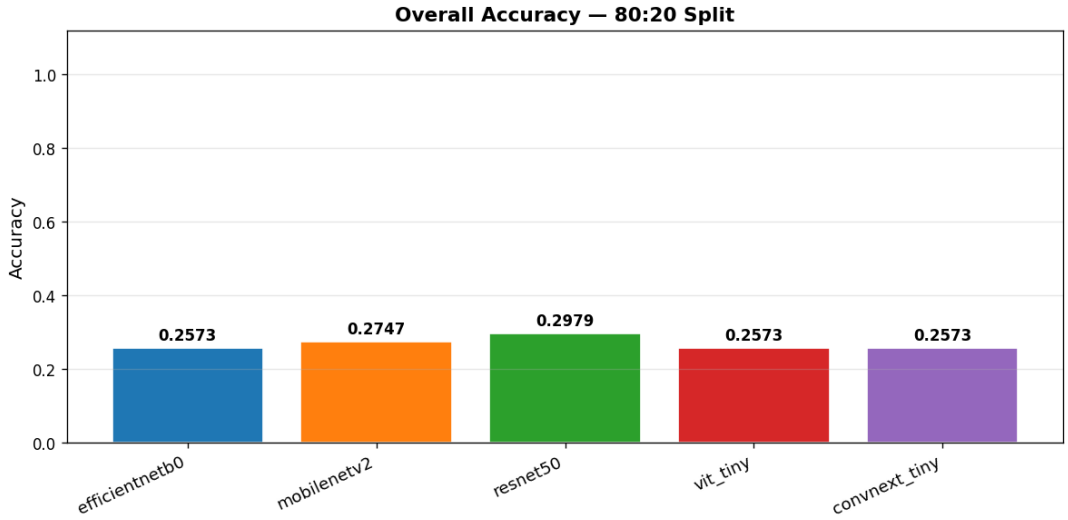

Figure 29. Overall accuracy comparison for the 80:20 split.

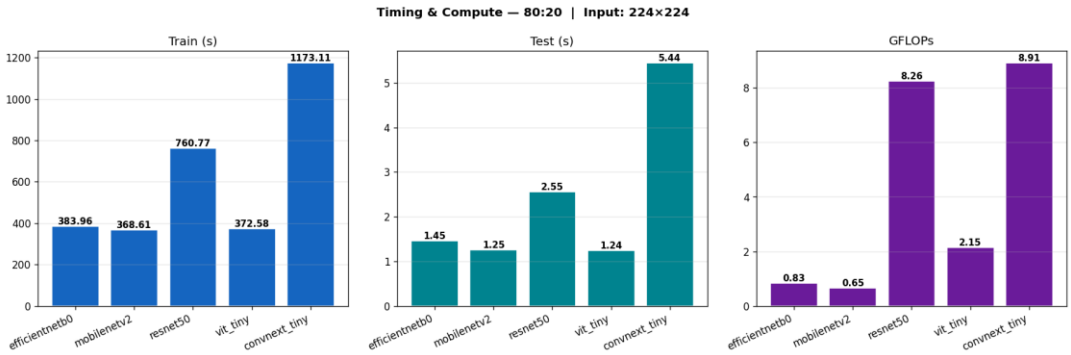

Figure 30. Training time, testing time, and GFLOPs comparison for the 80:20 split.

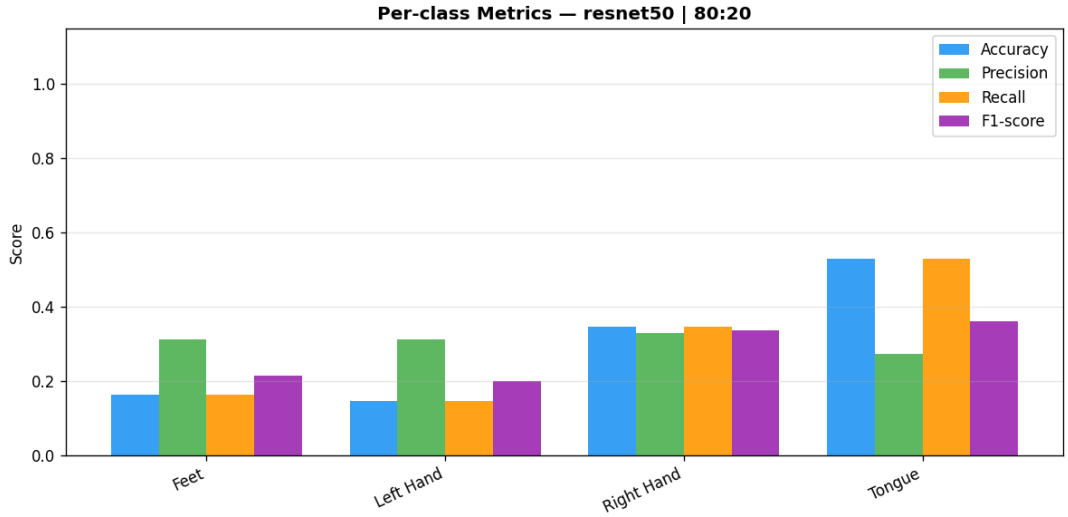

Figure 31. Per-class accuracy, precision, recall, and F1-score for the best-performing model in the 80:20 split (ResNet50).

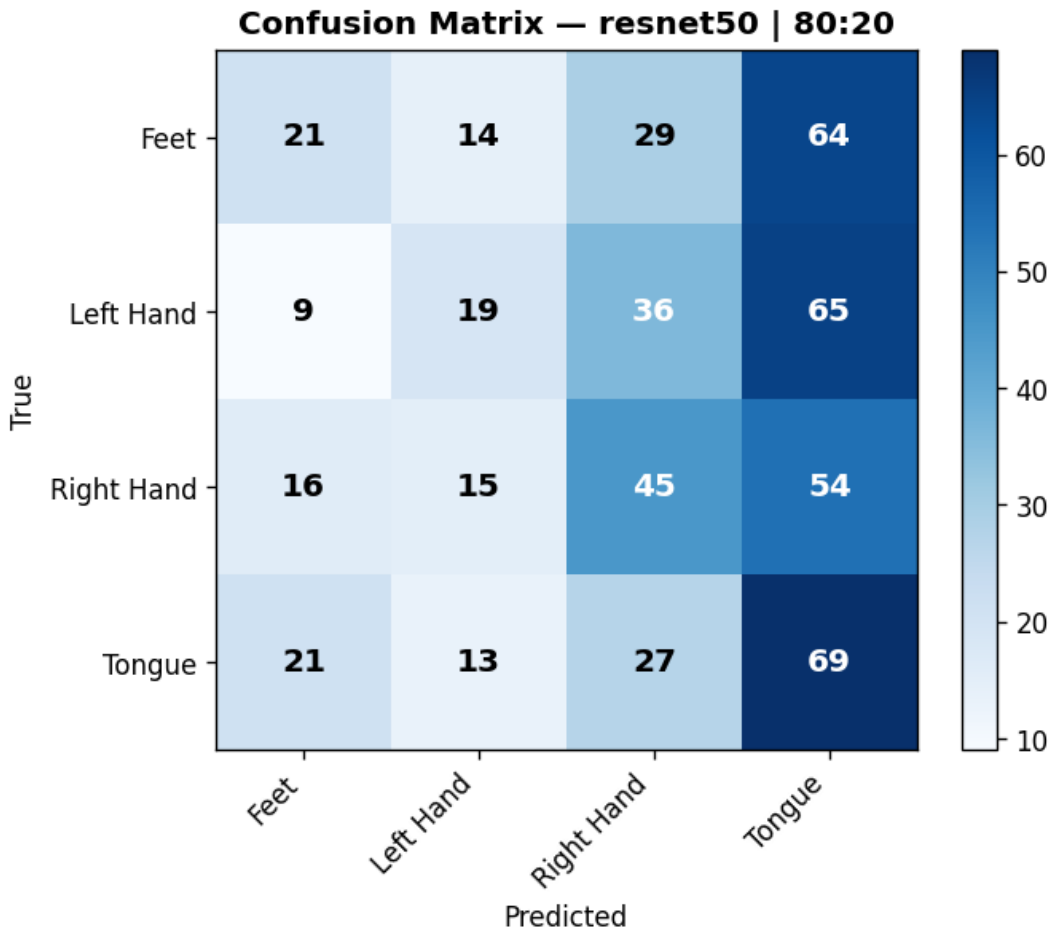

Figure 32. Confusion matrix for the best-performing model in the 80:20 split (ResNet50).

The best-performing supervised model for the 80:20 split is ResNet50, with accuracy = 0.2979, macro F1-score = 0.2784, and ROC-AUC = 0.5418. The best model for this split is ResNet50. The confusion matrix shows that several samples are incorrectly assigned to the Tongue class. The most visible errors are Left Hand -> Tongue, Feet -> Tongue, Right Hand -> Tongue, and Left Hand -> Right Hand.

11. Results for the 90:10 Split

For the 90:10 configuration, the dataset was divided into 2091 training images, 233 validation images, and 259 testing images. The table below reports only the supervised Task 2 metrics available in the notebook output for this split.

Model	Accuracy	Precision	Recall	F1	ROC-AUC	Train(s)	Test(s)	GFLOPs
ConvNeXt-Tiny	0.2857	0.1510	0.2846	0.1870	0.5294	1047.5	3.7950	8.9096
ViT-Tiny	0.2819	0.1415	0.2808	0.1879	0.5369	401.1	0.7230	2.1493
EfficientNet-B0	0.2432	0.2431	0.2430	0.2407	0.5244	559.8	0.8810	0.8277
ResNet50	0.2394	0.2263	0.2396	0.2279	0.4850	1274.5	1.3100	8.2634
MobileNetV2	0.2355	0.2370	0.2350	0.2274	0.5129	401.2	0.7710	0.6524

Table 10. Supervised Task 2 results for the 90:10 split.

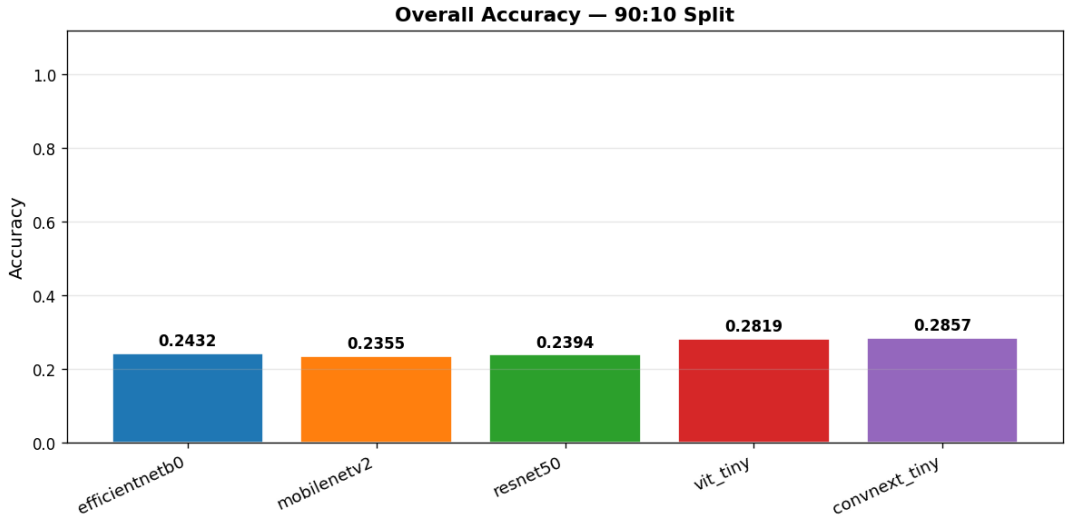

Figure 33. Overall accuracy comparison for the 90:10 split.

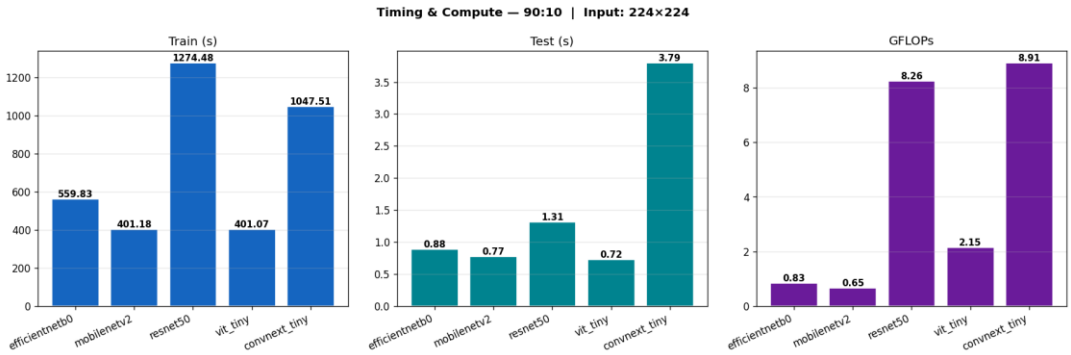

Figure 34. Training time, testing time, and GFLOPs comparison for the 90:10 split.

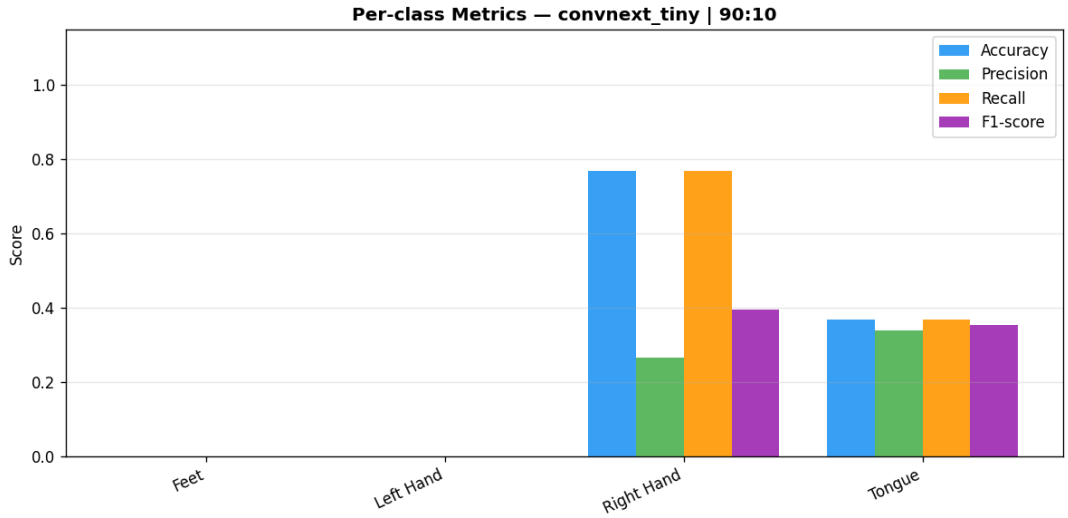

Figure 35. Per-class accuracy, precision, recall, and F1-score for the best-performing model in the 90:10 split (ConvNeXt-Tiny).

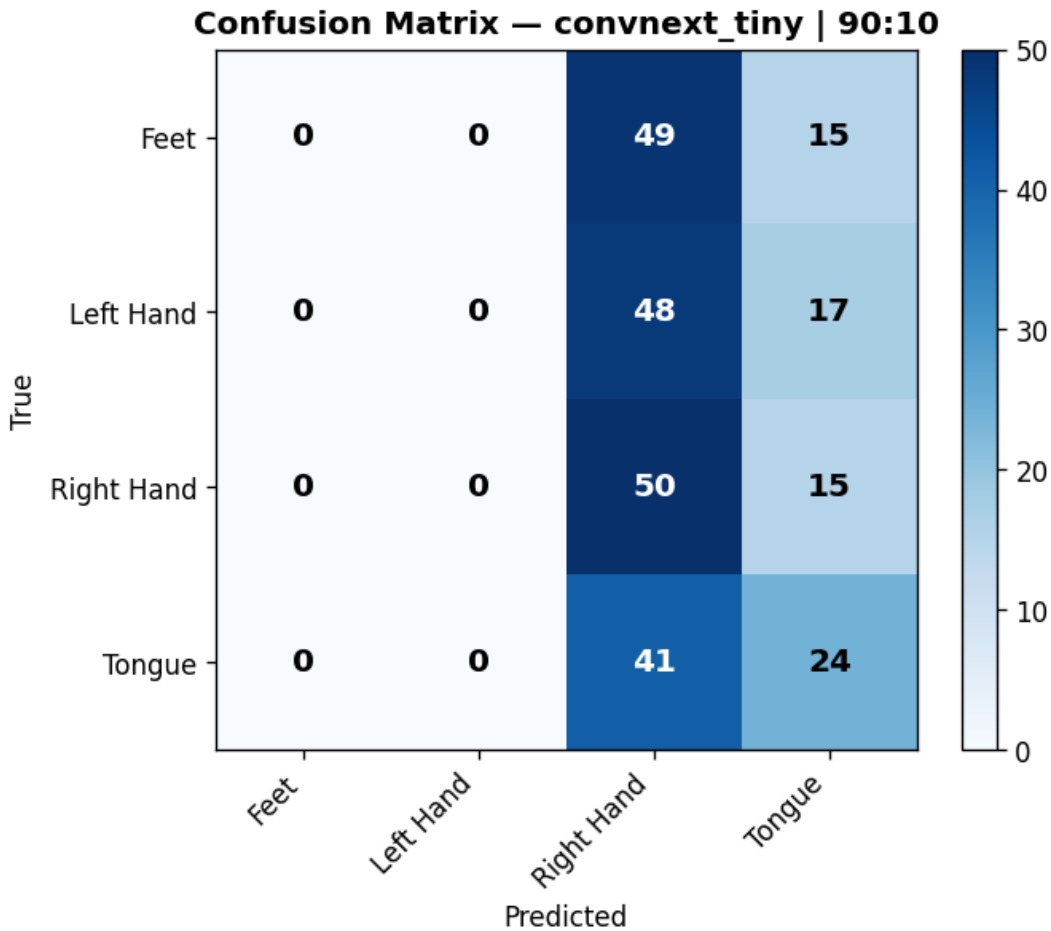

Figure 36. Confusion matrix for the best-performing model in the 90:10 split (ConvNeXt-Tiny).

The best-performing supervised model for the 90:10 split is ConvNeXt-Tiny, with accuracy = 0.2857, macro F1-score = 0.1870, and ROC-AUC = 0.5294. The best model for this split is ConvNeXt-Tiny. The confusion matrix shows a clear collapse toward the Right Hand class. The major errors are Feet -> Right Hand, Left Hand -> Right Hand, and Tongue -> Right Hand; Feet and Left Hand are not identified well.

12. Top-1 Supervised Baseline for SSL

Split	Model	Accuracy	Precision	Recall	F1	ROC-AUC	Train(s)	Test(s)	GFLOPs
80:20	ResNet50	0.2979	0.3068	0.2971	0.2784	0.5418	760.8	2.5540	8.2634

Table 11. Top-1 supervised result selected as the reference baseline for SSL.

From the analysis of all the supervised experiments done, the best supervised baseline experiment is the ResNet50 run on 80:20 dataset ratio. This experiment is chosen based on the fact that it achieved the highest supervised accuracy results in terms of Accuracy = 0.2979, Precision = 0.3068, Recall = 0.2971, F1-score = 0.2784, and ROC-AUC = 0.5418.

13. Required Deliverable Checklist

Required item	Status
Accuracy	Provided separately for all models and splits.
Per-class accuracy	Numbers of per-class accuracy from Task 2 provided for all best models by each split. Full per-class tables have not been printed.
Precision, Recall, F1-score	Macro results provided for all models and splits; per-class results provided via Task 6 per-class figures for best models.
ROC-AUC	Provided for all models and splits.
ROC curves	Not provided due to the lack of ROC plots in the code output; ROC-AUC scores only are printed.
Confusion matrix	Provided for best-supervised models found in Task 2 on each split.
Training and testing wall-clock time	Provided separately for all models and splits.
GFLOPs per inference	Provided for all models and splits with input 224 x 224.
Failure modes	Described separately for each split based on the confusion matrix.
Best supervised baseline for SSL	Described in the top-1 baseline section.

Table 12. Checklist against the instructor-required Task 2 deliverables.

14. Conclusion

This report presents the Task 2 supervised baselines are shown independently for each split used based solely on the information output from the uploaded notebooks. As we can see from the comparison of supervised baselines, the supervised classifiers exhibit low levels of accuracy overall, together with some amount of confusion between classes when it comes to spectrogram images. The best top-1 supervised baseline among those available is ResNet50 for the 80:20 split, which will be used as the baseline for SSL. Additional elements, which are not printed by the notebooks but are required, such as the ROC-curves and full tables for all models, are not included artificially.